

WATERWALKER · REQUIREMENTS

WaterWalker requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Values from WWK-CAL-001 v0.4 after the decisions of 2026-10-02 were carried into the design (hold-to-release brake, two-motor assist kit, torque-arm tabs on both rear forks); R10 and R13 wording for the new brake and tabs; no status changed
0.6	2026-10-02	Amish Chadha	R2 and R4 notes record Amish's decision of 2026-10-02 to accept them as not met on paper (WWK-DEC-001); no status changed
0.5	2026-10-02	Amish Chadha	Status from WWK-CAL-001 v0.3 for the constructable design (WWK-DDR-003); R11 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R10 restated for the positive lock pin; status updated from WWK-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3. Record Amish's decisions (WWK-DDR-001); redefine R1 (four jerrycans or two clay pots) and R11 (budget covers the unassisted first

prototype with mounting points); add R13 (assist-ready to SwapCell interface v0.3); add status from WWK-CAL-001

0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

WaterWalker requirements

Four of the thirteen requirements are not met on paper (R2, R3, R4 and R9), five are at risk and two are met, according to WWK-CAL-001 v0.4; R11 is over its value-engineering target and R13 cannot be verified at TRL 3. Targets are still desk proposals and must be revised from co-design sessions with the intended users (WWK-PRB-001) before the design is frozen. On 2026-09-25 Amish decided to build the first prototype without assist but with mounting points, keep the USD 450 figure, and carry four jerrycans or two clay pots (WWK-DDR-001); R1 and R11 were redefined to match, and R13 is new. Later the same day Amish accepted the remaining recommendations (WWK-DDR-002), including a positive parking lock pin, for which R10 is restated. On 2026-10-02 the design was made constructable (WWK-DDR-003): the parts added for construction made the carrier 2.2 kg heavier, which moved R2 and R4 from at risk to not met. Amish accepted the changes and R2 and R4 as not met on paper the same day, and decided on a hold-to-release brake and a two-motor assist kit; carried into the design, these add 0.7 kg, so R2 and R4 are now short by 1.2 N and 2.6 N (WWK-DEC-001). On 2026-10-01 Amish set budgets as value-engineering targets, so R11 is reported against its target rather than as met or not met.

Table 1. Requirements and status at TRL 3 (WWK-CAL-001).

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
R1	Carry a household's water in one trip	80 L of water in four 20 L jerrycans, or 40 L in two 20 L clay pots; rated gross payload 90 kg including containers (redefined 2026-09-25, WWK-DDR-001)	Cradle layout in the parametric model; later load test	At risk: two 380 mm pots fit with 12 mm to spare
R2	Easy to push on firm ground	Sustained push force 60 N or less at rated load on a firm, level dirt path	Rolling resistance calculation; later force gauge on a field path	Not met: 24.5 to 61.2 N (by 1.2 N); accepted on paper by Amish on 2026-10-02, with the limit confirmed with users in co-design (WWK-DEC-001)
R3	Usable on loose sand	Sustained push force 150 N or less at rated load on loose sand, with assist if fitted	Calculation from sinkage and rolling resistance; later field test	Not met: 241 to 367 N (no assist on the first prototype); 20 to 154 N with the two-motor kit
R4	Climb and descend slopes	Climb a 10 % grade on firm ground with a push force of 180 N or less for up to 100 m; descend a 10 % grade under control with the service brake	Force calculation; brake sizing	Not met: 182.6 N at the rough end (by 2.6 N); descent met on paper, and the hold-to-release brake alone holds the carrier on up to 12 % if the user lets go; accepted on paper by Amish on 2026-10-02, with the limit confirmed with users in co-design (WWK-DEC-001)
R5	Turn without lifting	180 degree turn within a 4.0 m diameter circle without lifting or skidding a wheel; reverse by pulling back	Geometry of the parametric model	Met: 3.66 m
R6	Fit narrow paths and gates	Overall width 900 mm or less	Parametric model	At risk: 898 mm
R7	Walk-in fit for all users	No step-over at the rear entry (50 mm or less); clear walking width 600 mm or more; hip bar height adjustable 850 to 1,050 mm; skirt guards on both rear wheels	Parametric model; later co-design fitting sessions	At risk: walking width 610 mm
R8	Easy loading	Containers lifted no higher than 450 mm above the ground; loadable from either side without removing parts	Parametric model	Met: 364 mm
R9	Light enough to handle empty	Empty mass 35 kg or less without assist, 42 kg or less with assist	Mass roll-up from the parametric model	Not met: 40.3 kg (49.6 kg with the two-motor kit); to be revisited with users in co-design (WWK-DDR-002)
R10				

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
	Safe braking and parking	Service brake on both rear wheels operable while walking; a positive parking lock (lock pin through a rear wheel) that needs no sustained hand force holds the rated gross mass on a 20 % grade; the lever latch serves short stops (restated 2026-09-25, WWK-DDR-002); a hold-to-release brake applies both rear brakes when the user lets go of the bail (decided 2026-10-02, WWK-DEC-001)	Brake force calculation; later parking and descent tests	At risk: lock pin held at both ends; tire friction 0.41 needed; latch 192 N; hold-to-release brake squeeze 98 N, hold 24 N (estimate)
R11	Affordable first prototype	Parts for the first prototype, without assist and with assist mounting points, against a value-engineering target of USD 450 (<code>project.yaml</code> budget, kept 2026-09-25; a hypothetical control target, not a limit, 2026-10-01). The budget excludes the optional assist kit, which is a later second prototype, and the SwapCell pack, which is priced once in the SwapCell BOM (redefined 2026-09-25, WWK-DDR-001)	Priced BOM	Over the target by USD 42: USD 492
R12	Repairable in a rural market	All wear parts (tires, tubes, spokes, bearings, cables, brake shoes) are standard 26 in bicycle parts; mild steel frame repairable by a local welder; only common hand and bicycle tools needed	Design review of the BOM; later review with a local mechanic	At risk: 100 mm drum hubs are less common than rim-brake hubs
R13	Assist-ready (new 2026-09-25)	The first prototype carries mounting points for a later assist kit built to SwapCell interface v0.3: a plate for a latch class V1 vehicle receiver above the tires, with the pack's back and lid faces open to air; rear dropouts that accept a 100 mm hub motor, with a torque-arm tab on each rear fork for the two-motor kit (2026-10-02); a boss for a push sensor at the hip bar. The receiver fits a 10 kΩ INTERLOCK coding resistor (item W), and the assist draws no more than the 15 A legacy-mode limit	Parametric model; later latch class V1 test	Not verifiable at TRL 3: geometry present, 7.8 A peak

■ Assumptions

- A 20 L jerrycan weighs about 1.1 kg empty and measures about 360 x 175 x 430 mm. A 20 L clay pot is assumed to be about 380 mm in diameter and 5 to 8 kg empty.
- Sustained push forces are set with reference to published manual-handling guidance, which suggests roughly 100 to 150 N for sustained pushing by women over long distances and more for short efforts. The limits must be confirmed with the users, including girls and older women.
- A household of five uses about 100 L per day (about 20 L per person).
- The first prototype has no assist (WWK-DDR-001), so R3 cannot be met by it; R3 stays as the target for the later assist prototype, which needs about 244 N of thrust (two hub motors, about 41 N·m each) to meet it on the

loosest sand (WWK-CAL-001 v0.4). Its value-engineering target is about USD 450 for the two-motor kit, pack excluded (Amish, 2026-10-02).

- R5 allows a larger circle than a compact rollator because the concept needs a long wheelbase to keep the swivelling front wheels clear of the cradle (WWK-PRC-001). A three-point turn is expected on paths narrower than 4 m.